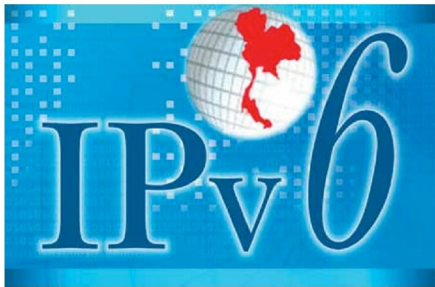


Are You Ready to Transit to IPv6?

What is IPv6, what if you don't transit to it and how to do it best—find out



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Internet protocol (IP) is a system of rules by which data is sent digitally across a network. It sends data as packets, called datagrams, over the network. Each of these data packets consists of a header and a payload. The header contains data such as the sender's and receiver's addresses, while payload contains the actual information.

IPv4 is the fourth and also the most widely deployed version of IP. It is this popularity that has ultimately led to the need for an updated IPv6 protocol.

As IPv4 increased in popularity and more devices used it, the Internet has almost run out of IPv4 address space. IPv4 uses 32-bit addresses, which gives a total of 2^{32} addresses. This means that it has a limited space of 4,294,967,296 addresses. The fact that this address space could exhaust soon was not

anticipated by the original designers.

What happened to IPv4?

Internet-enabled devices like smartphones and tablets are flooding the market. Almost all the smart devices released these days require an always-'on' Internet connection to run various applications and other functions. This always-'on' status causes the device to retain its IP address indefinitely, preventing others from getting it. This has caused the address uptake by service providers to escalate rapidly and the Internet to run out of IP address space.

Historically, inefficiencies plagued the initial address allocation, as the classful network allocation followed back in the eighties was inadequate to reflect reasonable usage. Even after the problem was discovered and a re-design of the addressing system using a classless network model done, it became clear that this would not prevent IPv4 address exhaustion, and a major

change to the Internet infrastructure was needed.

What if you don't transit to IPv6?

Exhaustion of IPv4 address was understood half a decade ago, but it did not seem an imminent threat so people ignored it. A slumping economy also had a hand in changing the focus to other issues. But now it has become an important concern.

There are workarounds like NAT and tunneling solutions that allow one to dodge the transition. But in such workarounds, in-line processing will add latency to the system, and sticking to an outdated protocol will increasingly degrade the performance. Firewalls will also create problem and the system security might go for a nose-dive.

Communities like Anonymous are waiting for the transition, to go about on a hacking spree. Firewalls won't recognise IPv6 traffic until these are programmed to do so—meaning legacy systems would allow IPv6 packets to go straight through them.

There are IPv6 botnet and malicious code packets already doing the rounds searching for unsuspecting victims. If an organisation does not require IPv6, it is recommended to block all IPv6 traffic until transition.

How to transition to IPv6?

Before you take the jump, be aware that transitioning to IPv6 is not as easy as toggling a switch. It will require running both protocols simultaneously—unless, of course, one decides to alienate all IPv4 customers. The time when IPv4 is not needed at all lies years away. But the security and performance implications of not switching to the new protocol are serious as well.